

DNN workload
(layer sizes, DAG)

IMC architecture: 128×128
CBs, c = CBs per tile

Packing

1 Architecture & packing

PD: density c (hardware-fixed)
PO: orientation (r,s) , $r \cdot s = c$
choose (r,s) : min PF
out: traffic graph (rates, phases), PF

traffic graph

sets PF

Mapping

2 Mapping (tile $\rightarrow$ node)

place at min CC; refine: max ES
s.t. $CC \leq CC_{min}$, $PL \leq PL(CC_{min})$
out: mapping π , PL, CC

sets π , PL

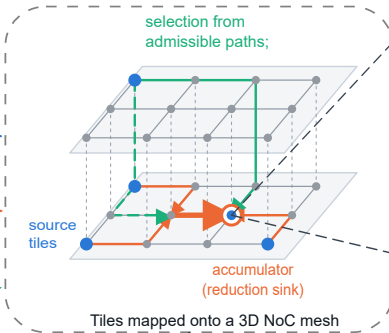
placement π

picks the path

Routing

3 Routing & selection (run time)

routing: odd-even balanced admissible paths $\{R\}$
selection: BL (1-hop buffer occupancy)
DP (multi-hop cost-to-go)



Average and tail delay(p99)

$PF = \max(PIL, PEL)$ — packing-fixed
 $PL = \max_{\ell} \Lambda_{\ell}$ — mapping-set
 $B = \max(PF, PL)$ — predicts delay

